SYDE 671 / ASSIGNMENT 3 / FALL 2026

Photo mosaics. *Manual to automatic.*

Recovering a shared perspective from overlapping photographs.

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Figure 1. City panorama: automatic mosaic from 3 photographs. The source images, feature matches, and manual comparison are documented below.

3 scenes
10 selected mosaic photographs

2 rectifications
Nearest neighbor and bilinear, implemented from scratch

6 mosaics
3 manual and 3 automatic estimates

Part A builds a projective image stitcher from manually chosen landmarks. Part B replaces those landmarks with Harris corners, adaptive non-maximal suppression, normalized patch descriptors, and four-point RANSAC. Both paths use the same homography solver, inverse warp, and feathered compositor.

Part A · Geometry

A.1 Source photographs

A.2 Recovering homographies

A.3 Warping and rectification

A.4 Manual mosaics

Part B · Correspondence

B.1 Harris corners and ANMS

B.2 Feature descriptors

B.3 Matching features

B.4 RANSAC and automatic mosaics

A.1

Source photographs

All photographs in this report were supplied by the author for this assignment. Three overlapping sets depict a basketball court, an indoor lounge, and a city view; two additional photographs show book covers for rectification. The intended mosaic model is camera rotation about a fixed center. Camera motion was not measured, and depth-dependent residuals are consistent with foreground parallax and departures from that model.

All coordinates use **(x, y)** pixel centers, with the origin at the top-left. Original files are preserved. The working images are antialiased and reduced to at most **1,400 pixels** on their longest side. Saved original-to-working matrices carry annotations into the resized image frame.

1. Basketball court

Three central frames of the basketball-court sweep (IMG_1242–1244) share the window wall and backboard. IMG_1243 is the reference.



Figure 2. IMG_1242. Working image: 1400 × 1050 pixels.



Figure 3. IMG_1243 · reference view. Working image: 1400 × 1050 pixels.



Figure 4. IMG_1244. Working image: 1400 × 1050 pixels.

The central three frames give a finite projective mosaic. The full six-image sweep crosses the projective horizon in this reference frame; IMG_1241, IMG_1245, and IMG_1246 are retained locally as supplemental inputs.

2. Lounge

Four overlapping lounge views, ordered IMG_1247, IMG_1248, IMG_1250, IMG_1249. IMG_1248 is the reference; the last two images are ordered by scene overlap rather than filename.



Figure 5. IMG_1247. Working image: 1400 × 1050 pixels.



Figure 6. IMG_1248 · reference view. Working image: 1400 × 1050 pixels.



Figure 7. IMG_1250. Working image: 1400 × 1050 pixels.



Figure 8. IMG_1249. Working image: 1400 × 1050 pixels.

3. City panorama

Three overlapping outdoor views (IMG_1251–1253), with IMG_1252 as the reference. Buildings provide static landmarks distributed across the lower half of the images; the sky is not used for manual annotations.



Figure 9. IMG_1251. Working image: 1400 × 1050 pixels.



Figure 10. IMG_1252 · reference view. Working image: 1400 × 1050 pixels.



Figure 11. IMG_1253. Working image: 1400 × 1050 pixels.

> **What the projective model assumes**
>
> A single homography describes a plane, or an arbitrary scene viewed by a camera rotating about a fixed center of projection. Handheld translation, changing depth, moving objects, and lens distortion can leave residual misalignment even when the fitted points agree. The source sets are retained so these limits can be inspected directly.

A.2 Recovering homographies

A homography maps a source point to a destination point up to scale: **$[u, v, 1]^T \sim H\,[x, y, 1]^T$**. Fixing $h_{33} = 1$ leaves eight unknowns. Each correspondence contributes two equations, so four nondegenerate pairs are the minimum. The manual mosaics use more than four pairs and solve an overdetermined system.

One correspondence

```
[ x  y  1  0  0  0  −ux  −uy ] h = u
[ 0  0  0  x  y  1  −vx  −vy ] h = v

h = [h11 h12 h13 h21 h22 h23 h31 h32]ᵀ
```

Stacking n correspondences gives $A \in \mathbb{R}^{2n\times 8}$ and $b \in \mathbb{R}^{2n}$. `computeH` centers each point set and scales its mean distance from the origin to $\sqrt{2}$, solves with `numpy.linalg.lstsq`, and transforms the solution back into pixel coordinates. Rank-deficient point sets are rejected. Normalization improves conditioning without changing the geometric model.

Denormalize

```
H = Tdestination⁻¹ Hnormalized Tsource; then H ← H / h33
```

The displayed matrices map the left source image to the right target image in **working-image coordinates**. They are rounded for readability; full-precision values and the annotated points are included in the measurement download. Errors below are residuals on the same landmarks used for fitting, not an independent estimate of photographic accuracy.

Scene	Source → target	Landmarks	Median residual (px)	Max residual (px)
Basketball court	IMG_1242 → IMG_1243	12	1.026	1.733
Basketball court	IMG_1243 → IMG_1244	12	1.093	2.155
Lounge	IMG_1247 → IMG_1248	12	2.176	6.816
Lounge	IMG_1248 → IMG_1250	11	2.258	6.002
Lounge	IMG_1250 → IMG_1249	11	2.055	4.528
City panorama	IMG_1251 → IMG_1252	10	0.688	1.048
City panorama	IMG_1252 → IMG_1253	11	0.393	0.717

A.2

Landmarks · Basketball court

IMG_1242 → IMG_1243

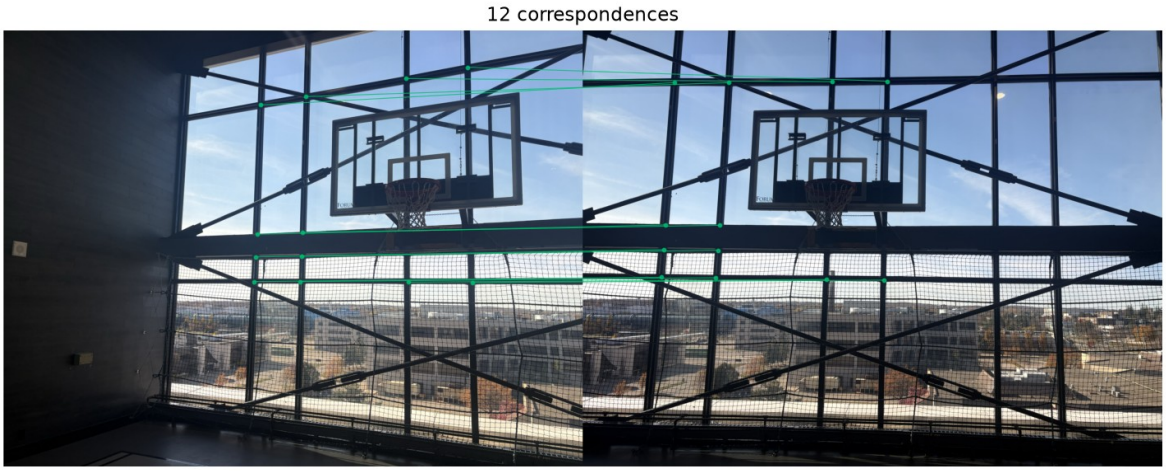


Figure 12. Basketball court: 12 manually selected corresponding landmarks, shown in green.

H · IMG_1242 → IMG_1243

$$\begin{bmatrix} 1.73134 & -0.0447851 & -771.757 \\ 0.323571 & 1.47209 & -298.927 \\ 0.000518232 & 1.9696e{-}05 & 1 \end{bmatrix}$$

IMG_1243 → IMG_1244



Figure 13. Basketball court: 12 manually selected corresponding landmarks, shown in green.

H · IMG_1243 → IMG_1244

$$\begin{bmatrix} 1.78429 & -0.0573305 & -853.753 \\ 0.345303 & 1.4712 & -288.963 \\ 0.000561144 & -3.36899e{-}05 & 1 \end{bmatrix}$$

A.2

Landmarks · Lounge

IMG_1247 → IMG_1248

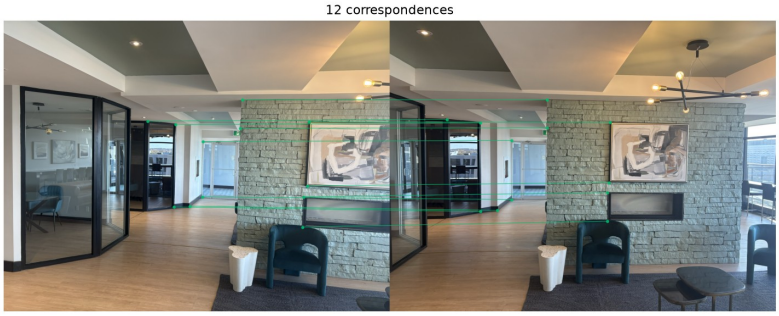


Figure 14. Lounge: 12 manually selected corresponding landmarks, shown in green.

H · IMG_1247 → IMG_1248

$$\begin{bmatrix} 1.52407 & 0.0363006 & -555.546 \\ 0.16491 & 1.3409 & -145.462 \\ 0.000395147 & -3.73692e{-}06 & 1 \end{bmatrix}$$

IMG_1248 → IMG_1250



Figure 15. Lounge: 11 manually selected corresponding landmarks, shown in green.

H · IMG_1248 → IMG_1250

$$\begin{bmatrix} 1.35703 & 0.0552419 & -453.912 \\ 0.129904 & 1.265 & -128.886 \\ 0.000239703 & 6.02928e{-}05 & 1 \end{bmatrix}$$

IMG_1250 → IMG_1249



Figure 16. Lounge: 11 manually selected corresponding landmarks, shown in green.

H · IMG_1250 → IMG_1249

$$\begin{bmatrix} 1.2708 & -0.0239527 & -274.474 \\ 0.0959206 & 1.10394 & -54.5096 \\ 0.000207818 & -5.90143e{-}05 & 1 \end{bmatrix}$$

A.2

Landmarks · City panorama

IMG_1251 → IMG_1252



Figure 17. City panorama: 10 manually selected corresponding landmarks, shown in green.

H · IMG_1251 → IMG_1252

$$\begin{bmatrix} 1.6121 & -0.0196081 & -610.387 \\ 0.245053 & 1.3938 & -232.292 \\ 0.000420283 & 3.57008e{-}05 & 1 \end{bmatrix}$$

IMG_1252 → IMG_1253



Figure 18. City panorama: 11 manually selected corresponding landmarks, shown in green.

H · IMG_1252 → IMG_1253

$$\begin{bmatrix} 1.85844 & -0.0382182 & -942.137 \\ 0.331829 & 1.51269 & -259.067 \\ 0.000635409 & -6.60971e{-}05 & 1 \end{bmatrix}$$

A.3

Inverse warping and interpolation

Forward splatting can leave holes. Instead, each output pixel is mapped backward by H^{-1} and sampled from the source. Automatic canvas bounds come from the transformed image corners. Rectification uses an explicit finite output rectangle. A separate validity mask distinguishes missing samples from valid black pixels.

Inverse mapping $[x', y', w']^\top = H^{-1}\,[u, v, 1]^\top;\quad (x, y) = (x'/w', y'/w')$

Nearest neighbor

Round the source coordinates to the nearest integer pixel and copy that value. The method keeps discrete source values but can produce jagged edges and blocky fine detail.

Sample

$$I_{out}(u,v) = I(\mathrm{round}(x), \mathrm{round}(y))$$

Bilinear interpolation

Let $x_0 = \mathrm{floor}(x)$, $y_0 = \mathrm{floor}(y)$, $a = x - x_0$, and $b = y - y_0$. Blend the four neighboring source pixels using the weights below. This smooths edge transitions at additional arithmetic cost.

Sample

$$(1-a)(1-b)\,I(x_0,y_0) + a(1-b)\,I(x_0+1,y_0) + (1-a)b\,I(x_0,y_0+1) + ab\,I(x_0+1,y_0+1)$$

Both methods are implemented directly in NumPy. They use the same closed domain of valid source pixel centers; the last row or column repeats its boundary neighbor for bilinear sampling. Neither method calls a library interpolation or projective-warp routine.

A.3 / 01

Letting Go front cover



Figure 19. Letting Go front cover: original with annotated corners, nearest-neighbor rectification, and bilinear rectification.



Figure 20. Identical title-region crops enlarged for inspection. Nearest-neighbor sampling shows sharper steps and broken fine strokes; bilinear sampling makes the edges smoother.

The four front-cover corners are mapped to (0, 0), (599, 0), (599, 899), and (0, 899). The output uses an approximate 2:3 width-to-height ratio because physical dimensions were not available. Straightening the quadrilateral therefore demonstrates projective rectification; it does not establish the object's true metric proportions. Physical cover curvature can remain.

H · working image → rectified image

$$\begin{bmatrix} 1.83648 & 0.20178 & -658.276 \\ 0.0188669 & 2.47747 & -1095.02 \\ -6.53106e{-}05 & 0.001049 & 1 \end{bmatrix}$$

A.3 / 02

The Magic of Thinking Big front cover



Figure 21. The Magic of Thinking Big front cover: original with annotated corners, nearest-neighbor rectification, and bilinear rectification.



Figure 22. Identical title-region crops enlarged for inspection. Nearest-neighbor sampling shows sharper steps and broken fine strokes; bilinear sampling makes the edges smoother.

The four front-cover corners are mapped to (0, 0), (599, 0), (599, 899), and (0, 899). The output uses an approximate 2:3 width-to-height ratio because physical dimensions were not available. Straightening the quadrilateral therefore demonstrates projective rectification; it does not establish the object's true metric proportions. Physical cover curvature can remain.

$$\text{H} \cdot \text{working image} \rightarrow \text{rectified image} = \begin{bmatrix} 0.82272 & 0.00169782 & -98.0427 \\ -0.109137 & 0.911069 & -94.3249 \\ -0.000251619 & 0.000140062 & 1 \end{bmatrix}$$

A.3

Measured speed and visible quality

Rectification	Output pixels	Nearest median (ms)	Bilinear median (ms)	Time ratio
Letting Go front cover	600 × 900	64.4	104.6	1.62×
The Magic of Thinking Big front cover	600 × 900	65.1	110.4	1.70×

One warmup per method; 9 timed runs each; alternate method order; warping only, excludes loading, H fitting and saving.

Environment: Python 3.12.14; macOS-15.5-arm64-arm-64bit. All timing samples are preserved in the rectification record.

Bilinear interpolation is slower in these measurements because each output sample combines four neighboring values. It produces more continuous letter contours in the close-ups, while nearest neighbor retains sharper intensity jumps and visible stair steps. Neither is a ground-truth reconstruction: a single photographed surface provides no reference image for PSNR or geometric accuracy.

The near-zero four-corner fit residual only confirms that the computed homography maps its own four constraints correctly. It does not measure corner annotation accuracy, surface planarity, or the unknown physical aspect ratio.

A.4

Blending into a shared frame

Pairwise homographies are composed into the selected reference frame. The reference image keeps the identity transform. All transformed source corners determine a common canvas, and every source is inverse-warped into it using bilinear interpolation.

Feathering $M(u,v) = \Sigma_i\ w_i(u,v)\ I_i(u,v)\ /\ \Sigma_i\ w_i(u,v)$

Each source receives a tent-shaped weight proportional to its distance from the nearest image boundary, normalized to one at its maximum. A positive one-pixel floor preserves isolated edge pixels. The image and its weights are warped together; invalid samples receive zero weight. The overlap is a normalized weighted average, so no image simply overwrites its neighbor.

Feathering reduces hard intensity seams, but it cannot repair a wrong transformation or remove two projections of a moving object. Transparent canvas regions indicate no source coverage. The full projective footprint is retained, making edge stretching and coverage visible.

Optional extension: preserve fine detail

A two-band experiment retains the same automatic homographies. It combines a mask-normalized Gaussian lowpass (σ = 16 working pixels) using feather weights, then takes the fine residual from the source with the largest feather weight. A Gaussian transition (σ = 1 pixel) softens the ownership boundary. This makes the lounge fixture clearer, but an ownership boundary can create a visible cut through another object. All original feather baselines are retained. The court additionally uses the source-preservation refinement described next.

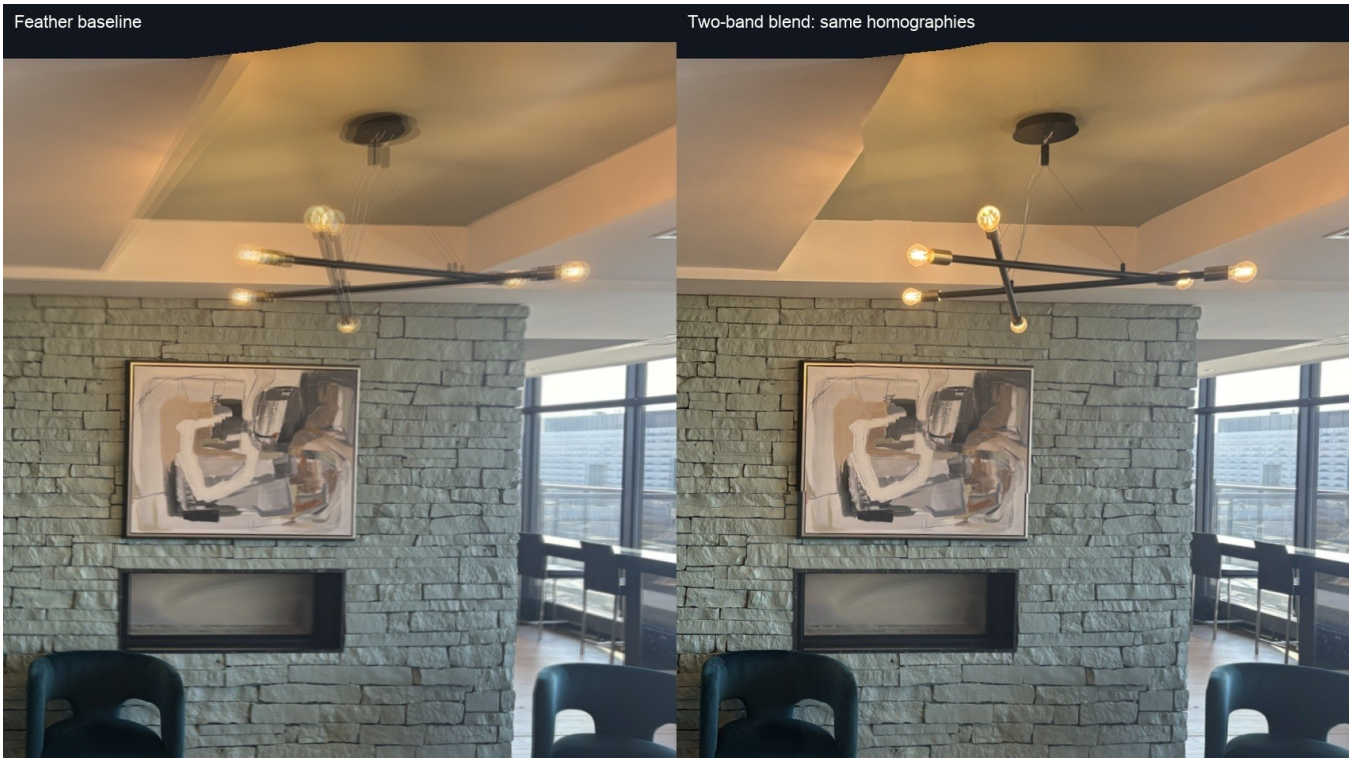


Figure 23. Optional two-band extension on the lounge, using identical automatic homographies and coverage. Detail ownership reduces duplicated chandelier edges and clarifies masonry while broad feathering softens exposure changes. Remaining overlap seams show that blending cannot repair parallax.

A.4 / 01

Basketball court



Figure 24. Basketball court: manual mosaic from 3 source images. Reference: IMG_1243. Canvas: 3072 × 1588 pixels.

Twelve manually observed window-frame landmarks constrain each adjacent pair. The displayed mosaic uses two-band blending and preserves the hoop from the reference photograph. The original feather-only result is retained below for comparison.

Measured shared-canvas warping and blending: 2.405 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

A.4 +

Keeping the hoop in one source view

The initial feathered mosaic doubled the backboard and rim because the foreground object occupies slightly different locations across the views. A source-selection refinement keeps this object intact while retaining the estimated homographies.

In the 1400-pixel-wide IMG_1243 reference frame, the manually specified rectangle x = 340–900, y = 140–560 uses that photograph exclusively across all frequency bands. A 72-pixel cosine transition lies outside the rectangle. Elsewhere, the two-band compositor retains sharper detail. The same physical region is applied to both manual and automatic geometry.

Protected pixels exactly equal the registered reference image before export, and the source-coverage mask is unchanged. This is a documented compositing intervention after geometric estimation; the automatic feather baseline remains visible below.



Figure 25. The complete automatic feather baseline, before the manually specified source-selection refinement.

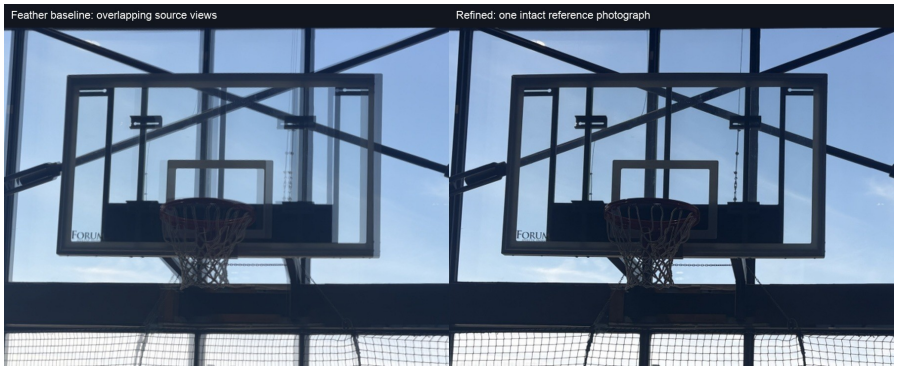


Figure 26. Automatic geometric estimate, identical before and after source selection. Left: feathering averages displaced edges. Right: the whole hoop, backboard, and net come from IMG_1243; no content is synthesized.

Unmodified feather baselines: Manual · Automatic · Both retain the same homographies as the displayed refinements.

A.4 / 02

Lounge



Figure 27. Lounge: manual mosaic from 4 source images. Reference: IMG_1248. Canvas: 2835 × 1553 pixels.

Manual points emphasize static door frames, artwork, the fireplace, and stone-wall details. The room is recognizable across all four views, with local doubling around the furniture and ceiling fixture.

Measured shared-canvas warping and blending: 2.946 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

A.4 / 03

City panorama



Figure 28. City panorama: manual mosaic from 3 source images. Reference: IMG_1252. Canvas: 3048 × 1687 pixels.

Ten and eleven visually selected landmarks constrain the two adjacent pairs. Refining ambiguous roof tips and using clear roof corners reduced the fitting residuals; the rooflines and chimney now remain sharp across most overlaps.

Measured shared-canvas warping and blending: 2.504 s. This is the single-run feather-baseline stage timing and excludes image loading, manual annotation, pairwise fitting, and optional blend refinements.

B.1

Harris corners and spatial coverage

The supplied Harris implementation detects single-scale corner candidates in a grayscale image. Its 20-pixel border exclusion leaves room for descriptor windows. Harris responses alone can cluster many detections around a textured object; adaptive non-maximal suppression spreads the selected features across the available image structure.

ANMS radius $r_i = \min_j \|x_i - x_j\|_2$, subject to $f_i < 0.9\, f_j$

For each candidate i, the radius is its distance to the nearest sufficiently stronger corner. The largest radii are retained, up to **500 features per image**. Candidates without a stronger suppressor receive a finite radius larger than the candidate cloud's diagonal; ties are resolved by response and input index. The implementation evaluates all candidate pairs in bounded-memory chunks. Descriptor validation can subsequently remove low-contrast patches.

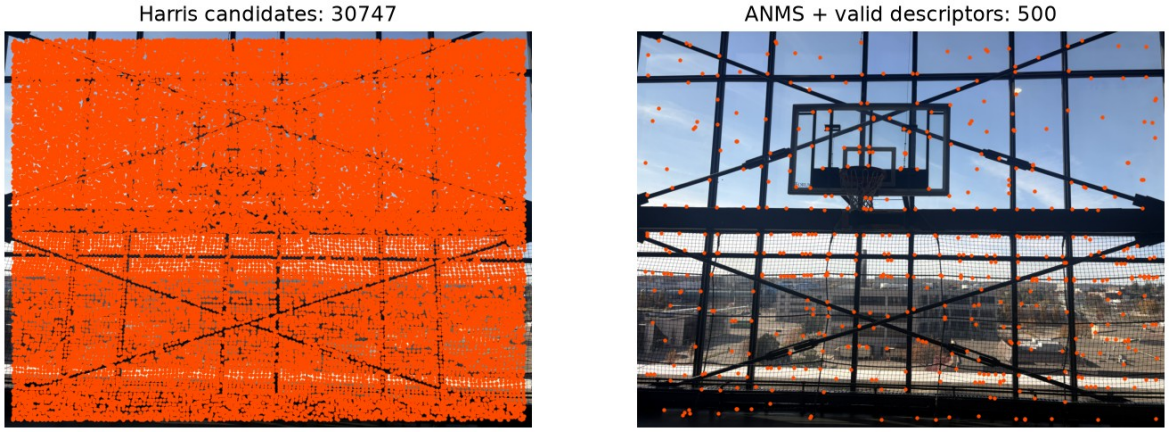


Figure 29. Basketball court, IMG_1243: all Harris candidates on the left; selected ANMS features with valid descriptors on the right.

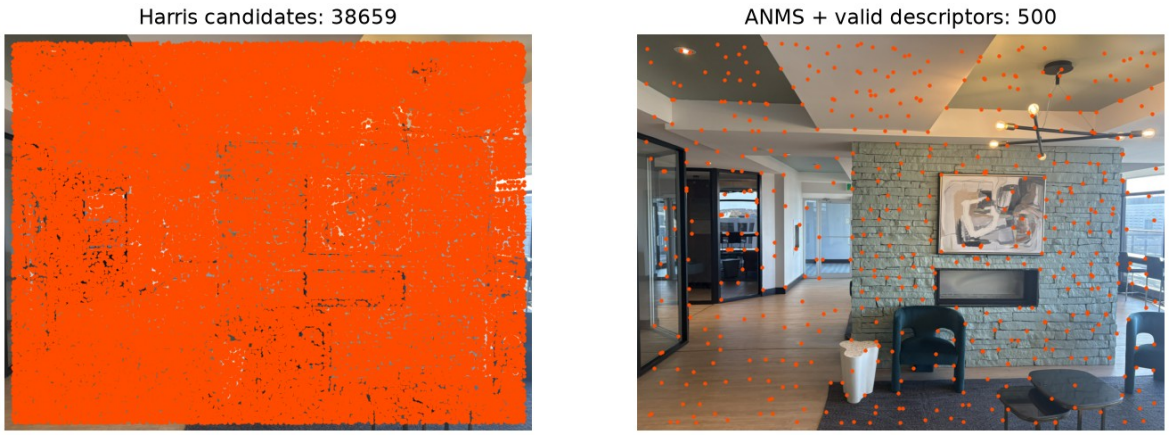


Figure 30. Lounge, IMG_1248: all Harris candidates on the left; selected ANMS features with valid descriptors on the right.

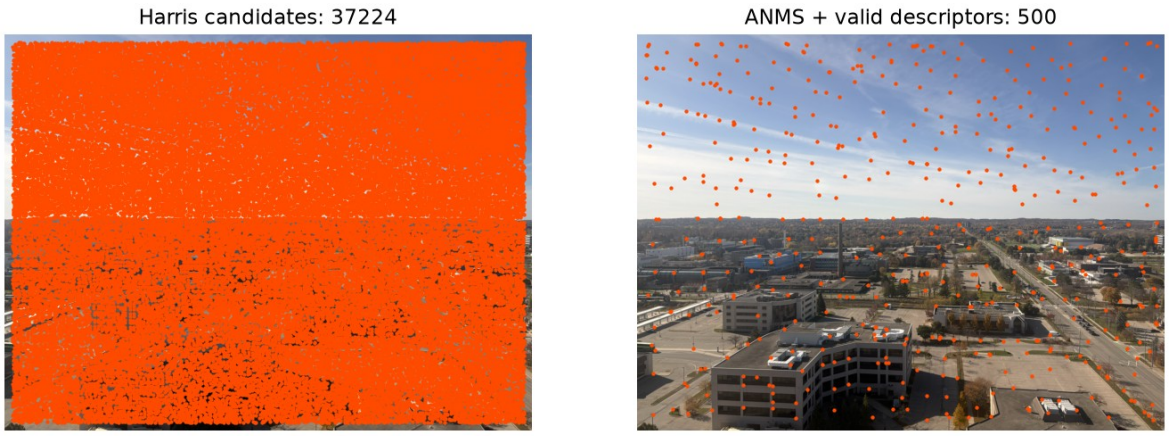


Figure 31. City panorama, IMG_1252: all Harris candidates on the left; selected ANMS features with valid descriptors on the right.

Scene	Image	Retained valid descriptors
Basketball court	IMG_1242	500
Basketball court	IMG_1243	500
Basketball court	IMG_1244	500
Lounge	IMG_1247	500
Lounge	IMG_1248	500
Lounge	IMG_1250	500
Lounge	IMG_1249	500
City panorama	IMG_1251	500
City panorama	IMG_1252	500
City panorama	IMG_1253	500

B.2

A descriptor from a larger neighborhood

A small patch is more useful when it summarizes a larger neighborhood. Each descriptor is an axis-aligned 8 × 8 grid sampled from a **40 × 40-pixel window**, after Gaussian smoothing with σ = 2 pixels. Grid samples are five pixels apart, at the centers of the eight bins. The sampling uses the same explicit four-neighbor bilinear formula, without a library interpolation call.

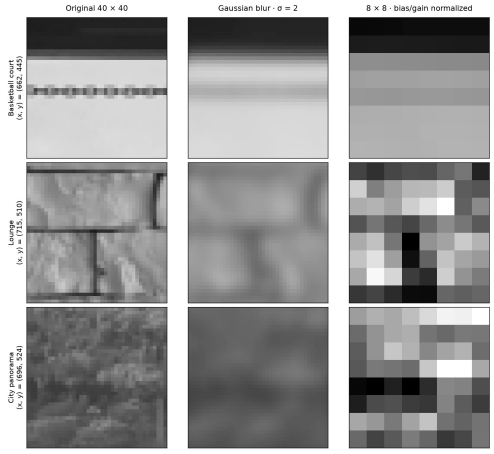


Figure 32. One actual matched feature from each reference image. Left: the original 40 × 40 grayscale neighborhood. Middle: the same neighborhood after Gaussian blur (σ = 2). Right: eight samples per axis, five pixels apart, normalized to zero mean and unit standard deviation. Normalized displays share the range −2 to +2; the exact 64 values are saved with the figure data.

Bias / gain normalization

$$d = (p - \text{mean}(p)) \,/\, \text{std}(p), \quad p \in \mathbb{R}^{64}$$

Mean subtraction removes an additive intensity offset; division by the standard deviation removes positive multiplicative gain. Nearly constant patches are discarded instead of dividing by a negligible standard deviation. This normalization does not make descriptors rotation-invariant or scale-invariant. The implementation follows the assignment's single-scale, axis-aligned simplification of MOPS.

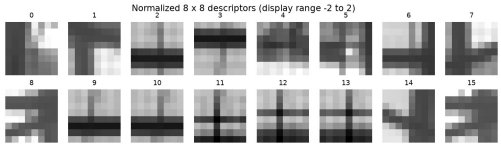


Figure 33. Basketball court: several normalized 8 × 8 descriptors from IMG_1243. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

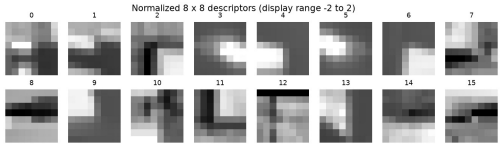


Figure 34. Lounge: several normalized 8 × 8 descriptors from IMG_1248. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

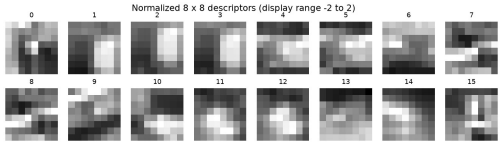


Figure 35. City panorama: several normalized 8 × 8 descriptors from IMG_1252. Display grayscale is clipped to −2 through +2 for a consistent visual scale.

B.3

Matching distinctive features

For every source descriptor, compute squared Euclidean distance to all target descriptors and find the nearest and second-nearest neighbors. A match must be clearly better than its closest alternative. This rejects many repeated or otherwise ambiguous patches before geometric fitting.

Ratio test $\rho = \|d_{source} - d_{nearest}\|_2^2 \,/\, \|d_{source} - d_{second}\|_2^2 < 0.5$

The configured threshold **0.5 applies to squared distances**, equivalent to 0.707 on ordinary Euclidean distances. Equal-distance ties and ambiguous 0/0 matches are rejected. The photographic pipeline uses the forward ratio test; reciprocal nearest-neighbor filtering is available in code but is not enabled for these results.

The ratio test measures descriptor distinctiveness, not geometric correctness. Green and red lines in the following figures show the later RANSAC decision: green matches support the estimated projective model, and red matches were rejected by its reprojection threshold.

Scene	Image pair	Ratio-test matches	Median accepted ratio
Basketball court	IMG_1242 → IMG_1243	75	0.280
Basketball court	IMG_1243 → IMG_1244	96	0.322
Lounge	IMG_1247 → IMG_1248	173	0.159
Lounge	IMG_1248 → IMG_1250	191	0.133
Lounge	IMG_1250 → IMG_1249	257	0.141
City panorama	IMG_1251 → IMG_1252	153	0.205
City panorama	IMG_1252 → IMG_1253	127	0.247

B.3

Correspondences · Basketball court

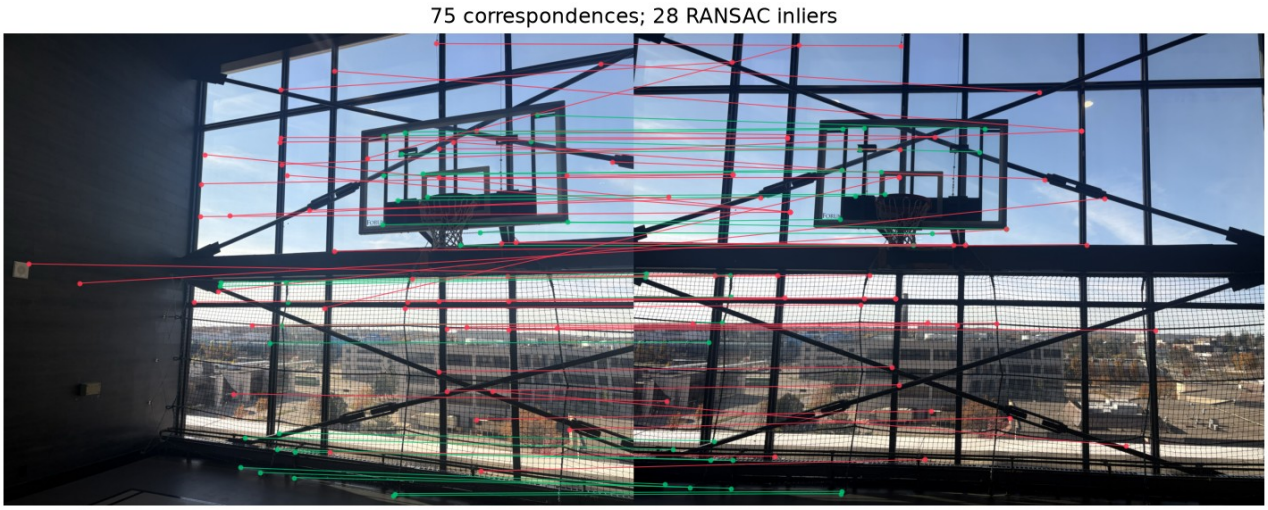


Figure 36. IMG_1242 → IMG_1243: 75 ratio-test matches; 28 RANSAC inliers (green) and 47 rejected matches (red).

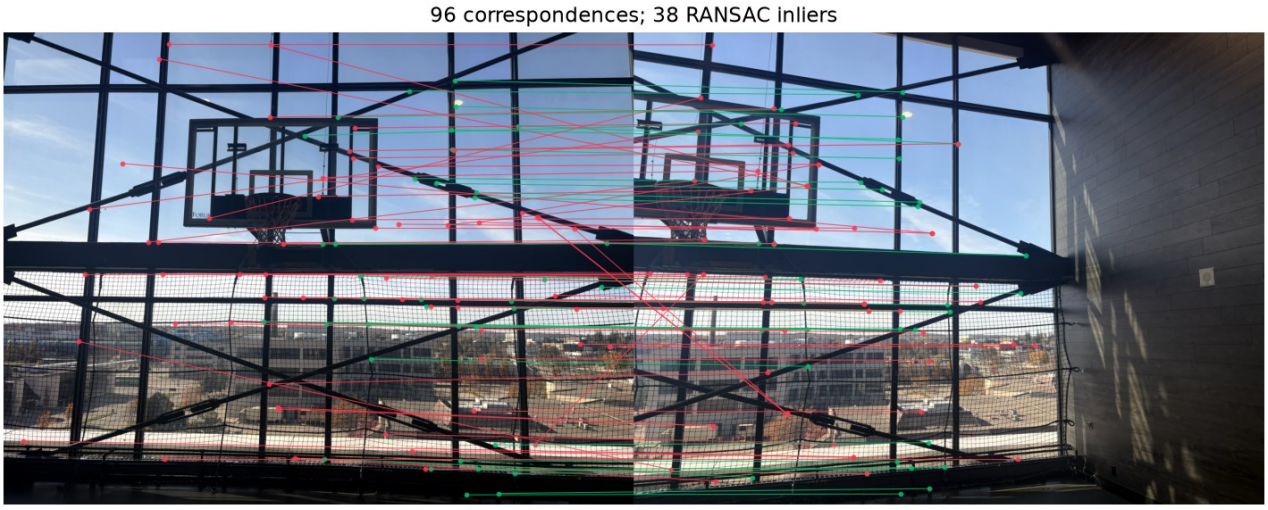


Figure 37. IMG_1243 → IMG_1244: 96 ratio-test matches; 38 RANSAC inliers (green) and 58 rejected matches (red).

B.3

Correspondences · Lounge

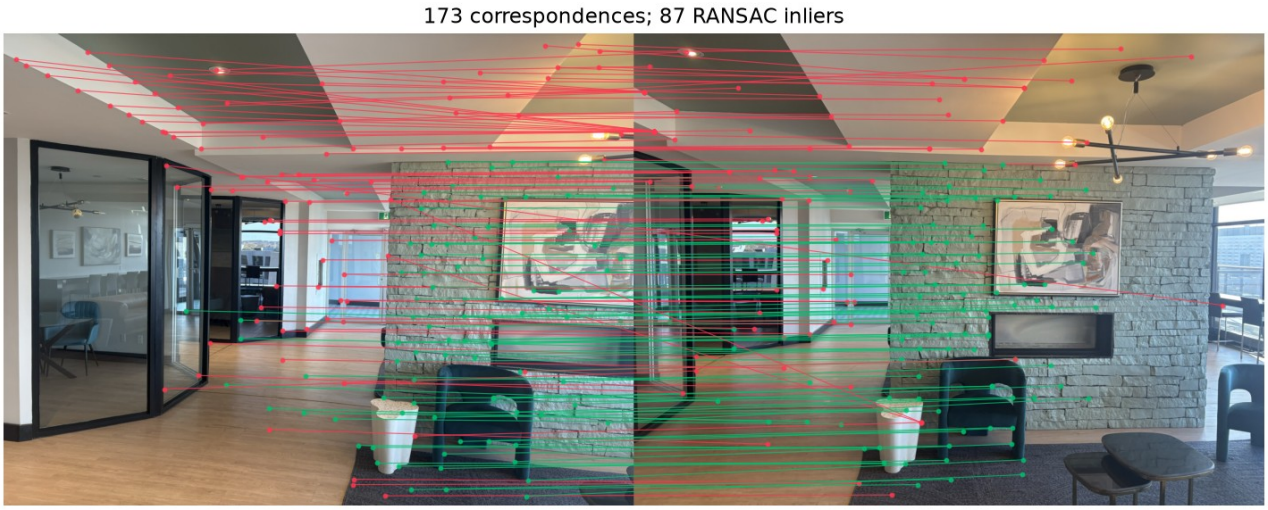


Figure 38. IMG_1247 → IMG_1248: 173 ratio-test matches; 87 RANSAC inliers (green) and 86 rejected matches (red).

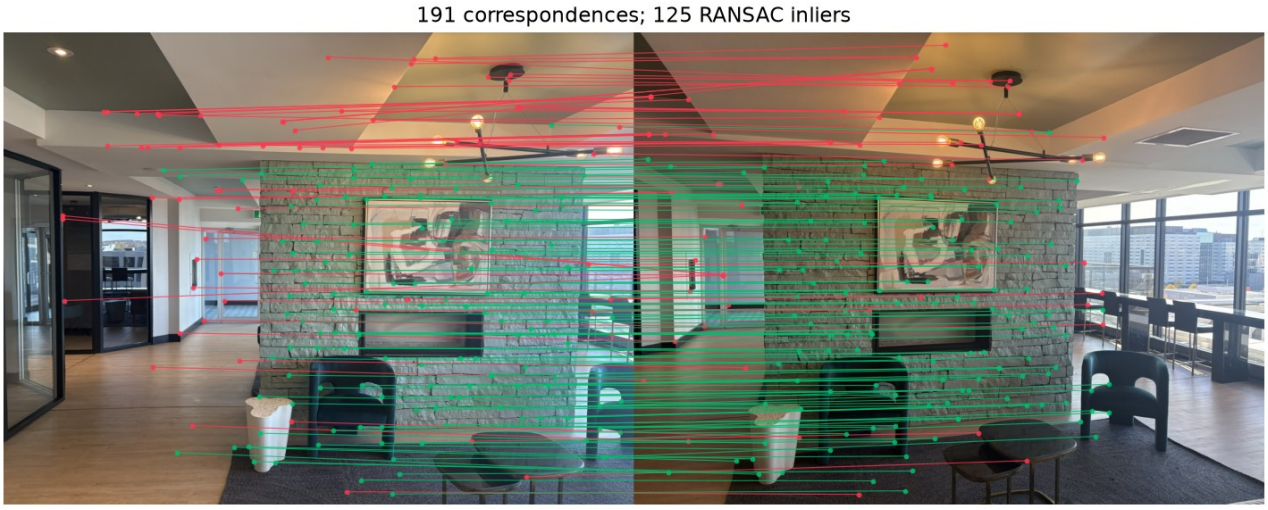


Figure 39. IMG_1248 → IMG_1250: 191 ratio-test matches; 125 RANSAC inliers (green) and 66 rejected matches (red).

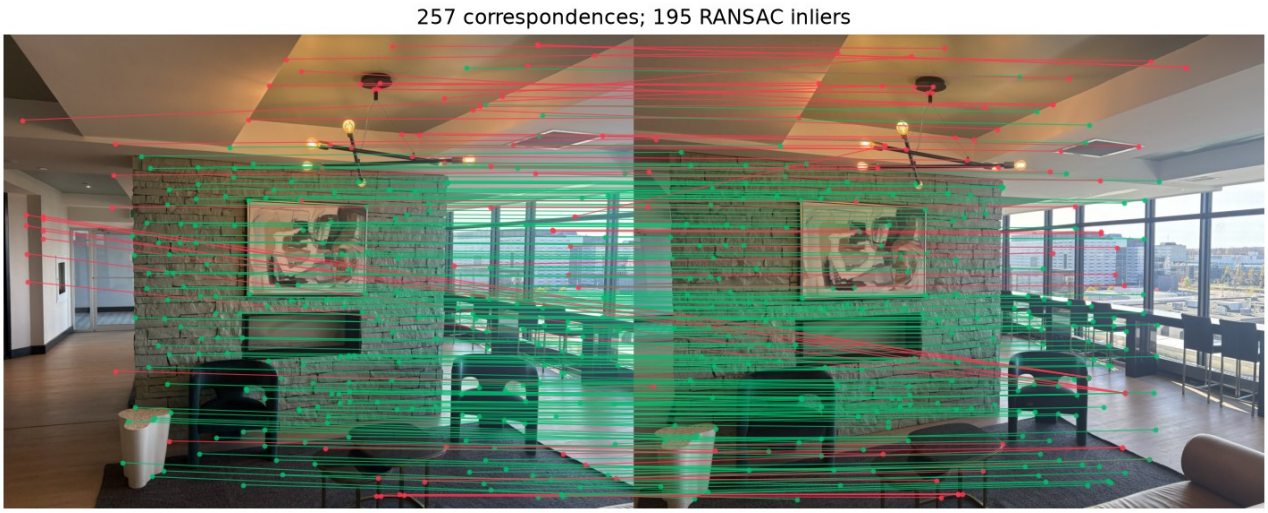


Figure 40. IMG_1250 → IMG_1249: 257 ratio-test matches; 195 RANSAC inliers (green) and 62 rejected matches (red).

B.3

Correspondences · City panorama

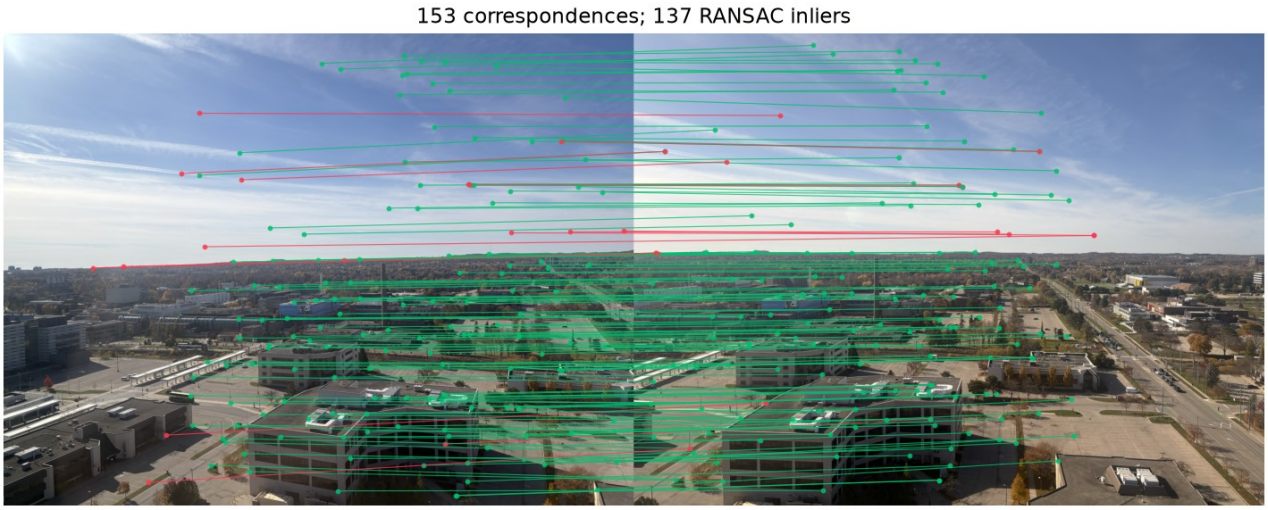


Figure 41. IMG_1251 → IMG_1252: 153 ratio-test matches; 137 RANSAC inliers (green) and 16 rejected matches (red).

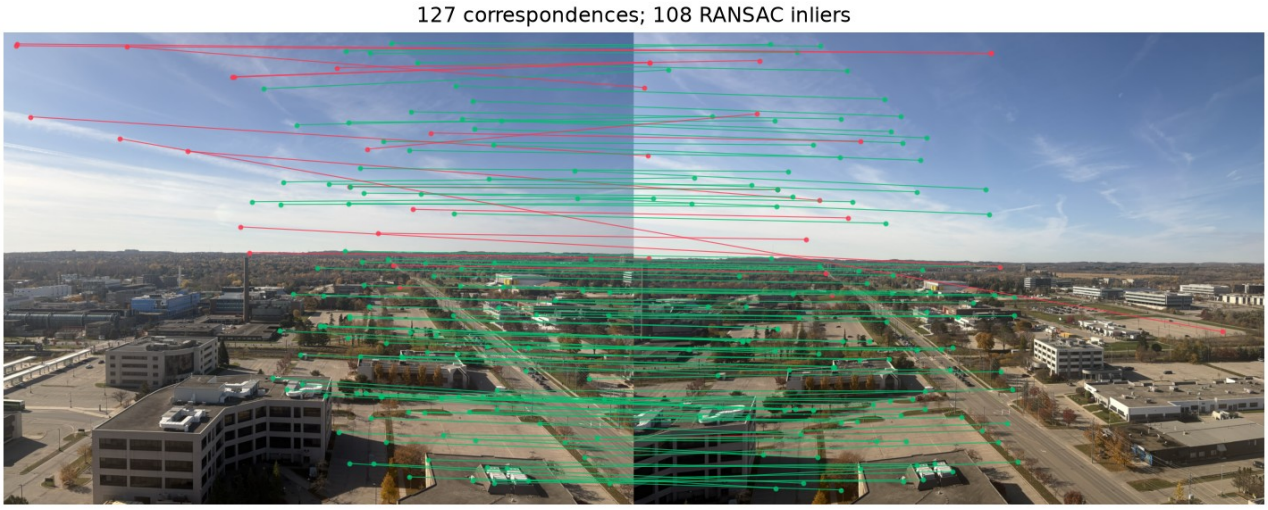


Figure 42. IMG_1252 → IMG_1253: 127 ratio-test matches; 108 RANSAC inliers (green) and 19 rejected matches (red).

B.4

A robust homography from noisy matches

Descriptor matches include outliers. A custom RANSAC loop repeatedly samples four distinct correspondences, fits a homography with `computeH`, and scores every match using forward reprojection distance. Degenerate samples are skipped. Hypotheses are ranked first by inlier count and then by median inlier error.

Inlier test

$$e_i = \|\text{project}(H\,[x_i, y_i, 1]^T) - [u_i, v_i]^T\|_2 \le 3 \text{ pixels}$$

The trial budget adapts to the observed inlier fraction using four-point sampling and a target confidence of 0.995, with a maximum of 5,000 attempts. The best consensus is refitted using all its inliers, with repeated refinement while support remains stable or improves. The final mask is recomputed from the returned homography.

Adaptive trial estimate

$$N = \text{ceil}\left(\frac{\log(1 - \text{confidence})}{\log(1 - \text{inlier_fraction}^4)}\right)$$

The run uses seed 671 plus the pair index. A scene pair must have at least 8 inliers and an inlier fraction of at least 20%. These are minimum support checks, not guarantees of a correct photograph-wide alignment.

Scene	Pair	Inliers / matches	Support	Trials	Median inlier error (px)	Match + fit (s)
Basketball court	IMG_1242 → IMG_1243	28 / 75	37.3%	271	1.125	0.059
Basketball court	IMG_1243 → IMG_1244	38 / 96	39.6%	403	1.224	0.120
Lounge	IMG_1247 → IMG_1248	87 / 173	50.3%	146	1.012	0.047
Lounge	IMG_1248 → IMG_1250	125 / 191	65.4%	81	1.186	0.041
Lounge	IMG_1250 → IMG_1249	195 / 257	75.9%	21	1.053	0.014
City panorama	IMG_1251 → IMG_1252	137 / 153	89.5%	18	1.096	0.014
City panorama	IMG_1252 → IMG_1253	108 / 127	85.0%	23	1.227	0.014

Residuals are forward errors on the fitted consensus, not held-out accuracy. Pair timing includes descriptor matching, RANSAC, and fitting; it excludes feature detection, image loading, and output generation. Trial counts include rejected degenerate samples.

B.4 / 01

Manual vs automatic · Basketball court

The source photographs, working resolution, reference image, bilinear sampler, and two-band blending with the same protected reference region are shared. The correspondence selection and resulting homographies differ.



Figure 43. Manual · Basketball court. Full projective canvas: 3072 × 1588 pixels.



Figure 44. Automatic · Basketball court. Full projective canvas: 3055 × 1557 pixels.

Both displayed results use the same two-band compositor and the same manually specified reference-source region around the hoop. The backboard, rim, and net therefore remain intact in both. Their sharpness is a compositing result and does not establish better homography accuracy; differences in the unprotected lower-window crop show the remaining geometric disagreements.



Figure 45. Manual (left) and automatic (right): the lower window and net outside the protected hoop rectangle, cropped at identical reference coordinates.

What remains imperfect

The backboard, net, window structure, and distant exterior lie at different depths. A single homography cannot align all of them if the camera translates. Source selection removes duplicated hoop edges locally; it does not correct parallax elsewhere, and ownership transitions can still leave seams.

B.4 / 02

Manual vs automatic · Lounge

The source photographs, working resolution, reference image, bilinear sampler, and the feathered compositor are shared. The correspondence selection and resulting homographies differ.



Figure 46. Manual · Lounge. Full projective canvas: 2835 × 1553 pixels.



Figure 47. Automatic · Lounge. Full projective canvas: 2771 × 1559 pixels.

The automatic result has sharper stone courses and a more consistent artwork frame in the common-coordinate detail. The manual result shows more widespread softness across the wall. Both retain displaced edges around objects nearer to the camera.



Figure 48. Manual and automatic estimates inspected at corresponding scene details.

What remains imperfect

Foreground furniture, the chandelier, and the corridor are not coplanar with the fireplace wall. Differences in viewpoint and exposure leave ghosting and broad intensity transitions that feathering cannot fully correct.

B.4 / 03

Manual vs automatic · City panorama

The source photographs, working resolution, reference image, bilinear sampler, and the feathered compositor are shared. The correspondence selection and resulting homographies differ.



Figure 49. Manual · City panorama. Full projective canvas: 3048 × 1687 pixels.



Figure 50. Automatic · City panorama. Full projective canvas: 3012 × 1659 pixels.

Both methods recover a coherent panorama from the same three photographs. The manual fits use 10 and 11 carefully observed landmarks, while the automatic fits retain 137/153 and 108/127 descriptor matches. These are fitting statistics, not independent accuracy measurements.

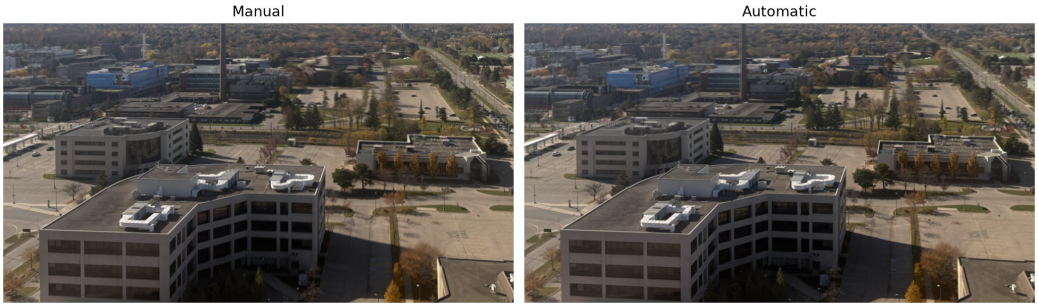


Figure 51. Manual and automatic estimates inspected at corresponding scene details.

What remains imperfect

Most landmarks are below the horizon, so the upper sky and outer image boundaries are extrapolated. Thin roof edges remain sensitive to small point-placement errors; the clouds are not assumed to be fixed geometric landmarks.

09

Validation and limitations

The photographic results test whether the components work together on real data, while known-answer tests check implementation behavior. A synthetic diagnostic is kept separate from the three original-photo mosaics and is not counted as a photographic submission example.

49 automated tests passed. Checks cover homography recovery and degenerate inputs, explicit inverse sampling and valid black pixels, ANMS and descriptors, ratio matching, RANSAC with outliers, transform chains, and project configuration. An independent degeneracy audit re-fitted all seven automatic photograph pairs from their saved matches; the corrected validation produced identical matrices, inlier masks, and trial counts. Five additional tests verify mask handling, detail ownership, and constant-field preservation in the optional two-band compositor. Two further tests verify that protected reference pixels remain exact and that the transition preserves coverage.

```
PYTHONPATH=src .venv/bin/python -m unittest discover -s tests -v
```

Known synthetic check	Measured result
Expected horizontal transform	-160 px
Matches / RANSAC inliers	260 / 250
Held-out maximum reprojection error	1.42e-13 px
Maximum mosaic intensity error	2.07e-13

This generated test has an exactly known translation and noiseless overlap. Its small numerical errors validate that controlled case; they do not predict the accuracy of handheld photographs.

Implementation boundary

The homography system, inverse mapping, nearest and bilinear sampling, feathering, ANMS, descriptor normalization, ratio matching, and RANSAC are implemented explicitly. NumPy provides linear algebra; SciPy provides Gaussian smoothing; the supplied Harris implementation is used as permitted. Pillow handles image I/O and orientation, and matplotlib produces diagnostic figures.

Limitations that affect interpretation

- A global homography cannot explain parallax, motion, or lens distortion across an arbitrary handheld scene.
- Axis-aligned, single-scale patches can fail under large rotation, scale changes, low texture, or repeated structures.
- Feathering averages disagreements; it does not remove geometric double edges or exposure differences.
- Consensus residuals and four-corner fitting residuals are not independent accuracy measurements.
- The rectified books use approximate display proportions rather than measured physical dimensions.

10

Reproducibility and references

The same configuration drives both manual and automatic processing. Input provenance, original-to-working transforms, correspondences, recovered matrices, RANSAC decisions, canvas bounds, and measured times are retained as JSON. The report builder reads those records directly.

Parameter	Value
Working longest side	1400 px
Maximum retained features	500
ANMS robust strength factor	0.9
Descriptor window / grid	40 × 40 / 8 × 8; 5-pixel sample spacing
Descriptor Gaussian σ	2 pixels
Squared-distance ratio threshold	0.5
RANSAC forward-error threshold	3.0 working-image pixels
RANSAC seed	671 + pair index
Baseline interpolation / blending	Bilinear / distance-to-boundary feathering

Reproduce the experiment

```
python3 -m venv .venv
.venv/bin/pip install -r requirements-lock.txt
PYTHONPATH=src .venv/bin/python -m unittest discover -s tests -v
.venv/bin/python run_project.py --mode all
.venv/bin/python run_rectifications.py
.venv/bin/python run_blending_experiment.py
.venv/bin/python run_gym_refinement.py
.venv/bin/python make_report_assets.py
.venv/bin/python run_synthetic.py
.venv/bin/python build_report.py
```

Run from the assignment folder. The input archive and configuration preserve the original-photo annotations; figure and report generation use the recorded results. Hardware and concurrent activity can change timings.

References

1. SYDE 671, Assignment 3: *[Auto]Stitching Photo Mosaics*. Course handout; adapted from Berkeley CS180 Assignment 3.
2. Brown, M., Szeliski, R., and Winder, S. (2005). *Multi-Image Matching using Multi-Scale Oriented Patches*. CVPR. The implementation follows the assignment's simplified single-scale, axis-aligned variant.
3. Assignment-supplied `harris.py`: Harris candidate detection and distance helper.

APX

Automatic homography matrices

These full-image pairwise transformations map source working coordinates into target working coordinates. The JSON download retains full precision and all image-to-reference compositions.

Basketball court · IMG_1242 → IMG_1243

```
1.66912  −0.0481914  −743.124
0.302618   1.44275  −285.728
0.000480189   9.73834e−06   1
```

Basketball court · IMG_1243 → IMG_1244

```
1.75943  −0.0620795  −835.532
0.332926   1.47492  −286.537
0.000542659  −2.53564e−05   1
```

Lounge · IMG_1247 → IMG_1248

```
1.39974   0.0190938  −492.918
0.11684   1.22941  −92.1788
0.000302281  −1.80218e−05   1
```

Lounge · IMG_1248 → IMG_1250

```
1.37391   0.00845299  −452.899
0.126847   1.22117  −110.258
0.000262928  −2.65058e−08   1
```

Lounge · IMG_1250 → IMG_1249

```
1.26089   0.00617285  −278.296
0.0834213   1.14026  −59.5377
0.000183846  −6.76548e−06   1
```

City panorama · IMG_1251 → IMG_1252

```
1.5937  −0.0133569  −607.481
0.244937   1.37944  −231.215
0.000415361   2.38386e−05   1
```

City panorama · IMG_1252 → IMG_1253

```
1.8885  −0.00571289  −981.234
0.336389   1.5636  −279.178
0.000637776  −3.16932e−05   1
```